

ful enough to playback or capture HD video, and they should be able to rapidly process the camera images at higher resolutions, in a wide range of lighting conditions. Many of the particular chips have neuromorphic designs to support machine learning applications, which are mostly used to ensure that your Instagram feed is all glammed up, or artificially deteriorate the quality of the images if your feed is on fleek. Let's take a peek inside the SnapDragon 855, which comes from the most widely used family.

On the SoCs are a number of chips. It has a number of CPUs arranged in complexes, a GPU, a modem, a satellite navigation module of a particular provider (Galileo, GLONASS, GPS and so on), and image processing chips to make the most of the information captured by the camera sensors. All the five families of SoCs mentioned above collaborate with ARM for their designs, and use the Arm Instruction Set. All the chips use at least one ARM Cortex core, with the exception of Apple. Let us peek inside an SoC to see how all these components are put together.

At the bottom left corner, the block labelled as the "Kryo 485" actually is a complex of eight cores, made up of four Kryo 485 Gold and Kryo 485 Silver units, which is a semi custom design used by Qualcomm. Each core has its own level 2 cache and a shared level 3 cache. On the top left of the SoC is the Qualcomm designed Adreno 640 GPU, which actually has a lower clock speed but promises 20 percent faster rendering over the previous generation, Adreno 630. The Hexagon 690 is the DSP, or digital signal processing unit, and is actually larger than the processor complex or the GPU. This is the part that ensures that you enjoy crystal clear voice conversations, and is actually one of the most important components on the phone. The Hexagon has a dedicated ML inferencing unit, which is useful for voice assistants, among other applications.

There are some smart design choices that companies can make in their SoC designs. The particular way these various components are integrated together can have different

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Flops

Stands for floating point operations per second, and is a measure of computing performance, and is more accurate for representing the power of a computer for certain applications, mostly scientific.

Flush

Short form for pipeline flush, which the CPU does when it cannot guarantee that it can correctly process the instruction pipeline in the next clock cycle. The pipeline is cleared to ensure the desired calculations are carried out.

Foundry

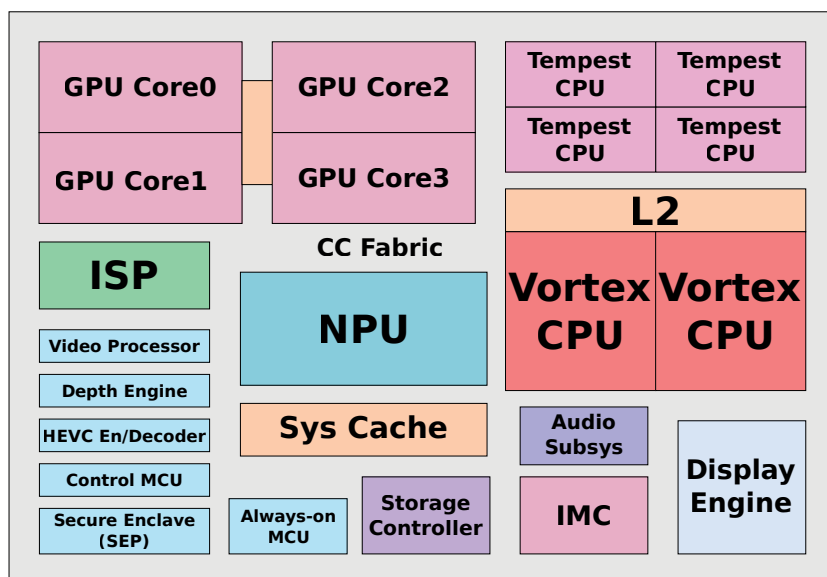
Also known as fab, which is short for a semiconductor fabrication plant.

FPGA

Field-programmable gate array are specialised ICs that can be programmed by the end users after the manufacturing. FPGAs allow users to tweak microprocessors for unique requirements.

benefits in terms of performance and power usage. These design choices include various generations or versions of cores, different voltage planes for delivering the requisite amount of power depending on applications, the number of rails routing the power, different clock speeds for the individual cores, dedicated units for AI, how closely or widely the cores are clustered, the number of cores, the amount of cache memory assigned to each core, the levels of cache memory used, the use of memory compression, how many channels the memory controller supports and so on. At times, a manufacturer may choose to pack in more custom components onto the SoC, instead of going for off the shelf components. Samsung, Huawei and Apple tend to do this the most.

In its A12 chip, Apple uses two large Vortex cores and four small Tempest cores. As compared to the previous generation, Apple claims the Vortex cores are 40 percent more efficient, while the Tempest cores are 50 percent more efficient. If you are spotting a pattern here, yes, reducing the energy footprint is one of the primary concerns of SoC design. As a general rule, the smaller the core, the lower the energy footprint. There are a lot of components in the Apple SoC, which is one of the most tightly packed together SoC in the market. The motherboard is actually folded over, allowing for components on both



The Apple A12 Block Diagram